

Intermediate Microeconomics

Week 12b: Uncertainty and Risk Aversion (II)

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The State-Preference Approach

- The approach taken in this chapter up to this point has not used the basic model of utility-maximization subject to a budget constraint.
- We will develop new techniques to incorporate the standard choice-theoretic framework

States of the World

- Outcomes of any random event can be categorized into a number of states of the world
 - “good times” or “bad times”
- Contingent commodities are goods delivered only if a particular state of the world occurs
 - “\$1 in good times” or “\$1 in bad times”
- It is conceivable that an individual could purchase a contingent commodity
 - buy a promise that someone will pay you \$1 if tomorrow turns out to be good times
 - this good will probably cost less than \$1

Utility Analysis

- Assume that there are two contingent goods
 - wealth in good times (W_g) and wealth in bad times (W_b)
 - individual believes the probability that good times will occur is π

Utility Analysis

- The expected utility associated with these two contingent goods is

$$V(W_g, W_b) = \pi U(W_g) + (1 - \pi) U(W_b)$$

- This is the value that the individual wants to maximize given his initial wealth (W)

Prices of Contingent Commodities

- Assume that the person can buy \$1 of wealth in good times for p_g and \$1 of wealth in bad times for p_b

- His budget constraint is

$$W = p_g W_g + p_b W_b$$

- The price ratio p_g/p_b shows how this person can trade dollars of wealth in good times for dollars in bad times

Fair Markets for Contingent Goods

- If markets for contingent wealth claims are well-developed and there is general agreement about π , prices for these goods will be actuarially fair
 - that is, they will equal the underlying probabilities:

$$p_g = \pi \text{ and } p_b = (1 - \pi)$$

- The price ratio will reflect the odds in favor of good times

$$\frac{p_g}{p_b} = \frac{\pi}{1 - \pi}$$

Risk Aversion

- If contingent claims markets are fair, a utility-maximizing individual will opt for a situation in which $W_g = W_b$
- he will arrange matters so that the wealth obtained is the same no matter what state occurs
- can also show the math...

Risk Aversion

- Maximization of utility subject to a budget constraint requires that

$$MRS = \frac{\partial V / \partial W_g}{\partial V / \partial W_b} = \frac{\pi U'(W_g)}{(1-\pi)U'(W_b)} = \frac{p_g}{p_b}$$

- If markets for contingent claims are fair, $\frac{p_g}{p_b} = \frac{\pi}{1-\pi}$

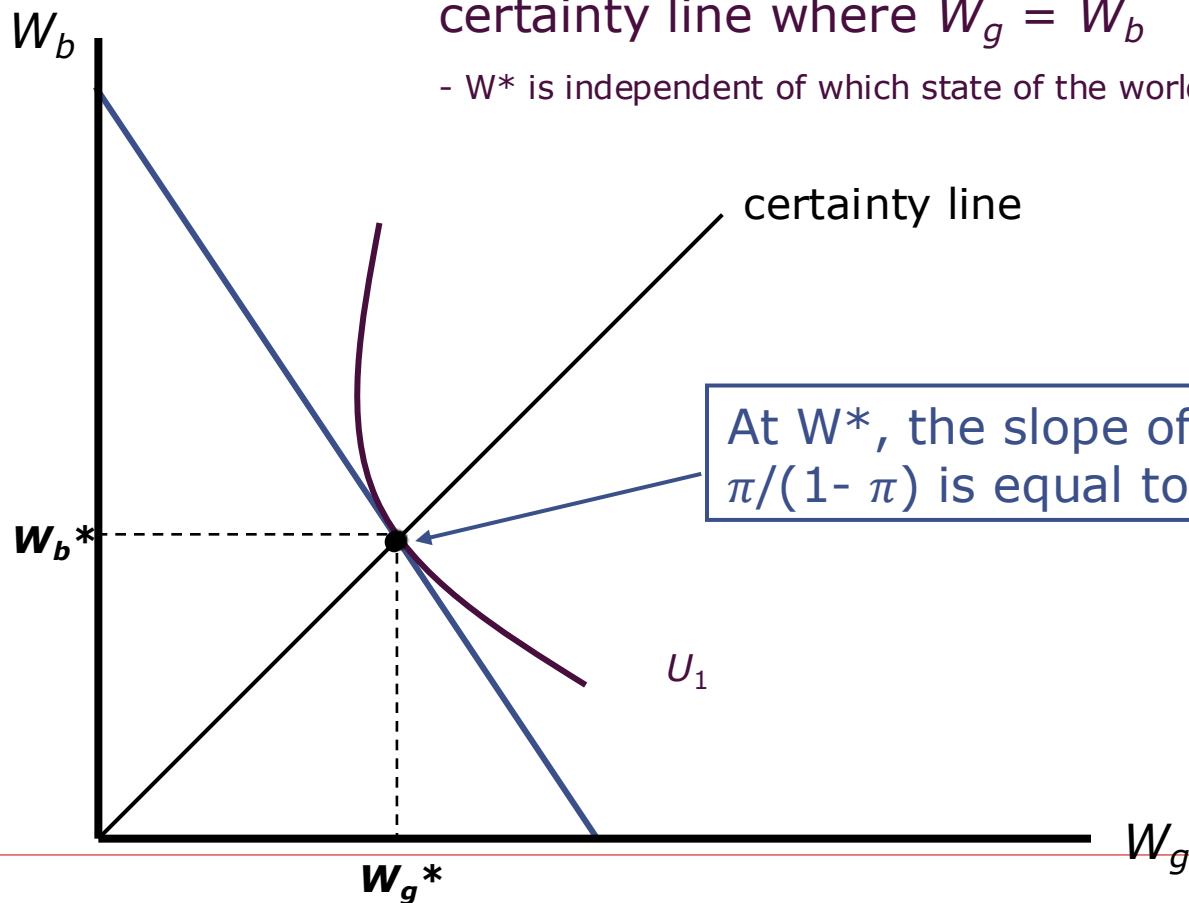
$$\frac{U'(W_g)}{U'(W_b)} = 1$$

$$W_g = W_b$$

Risk Aversion

The individual maximizes utility on the certainty line where $W_g = W_b$

- W^* is independent of which state of the world occurs

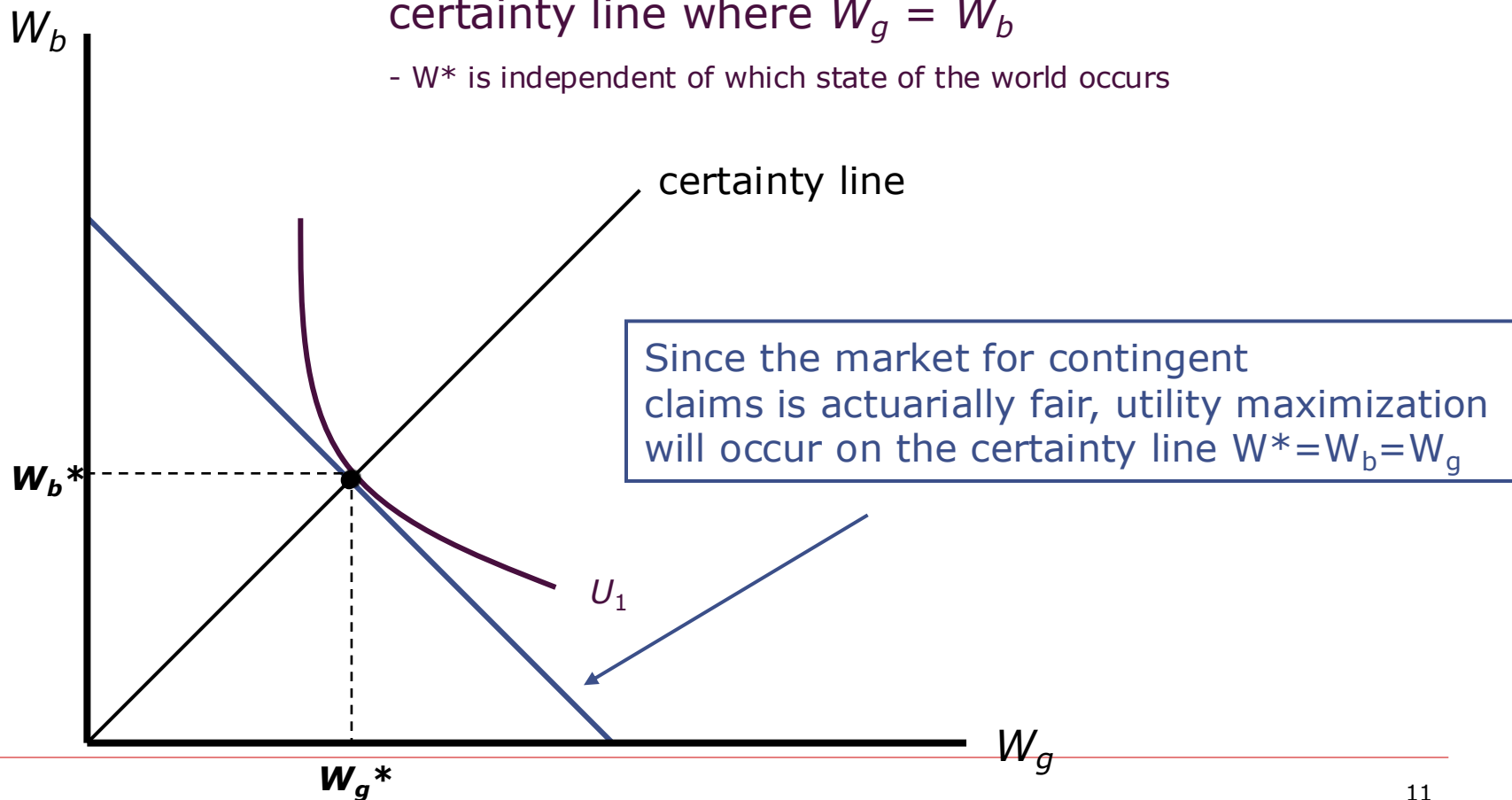


At W^* , the slope of the indifference curve $\pi/(1-\pi)$ is equal to the price ratio p_g/p_b

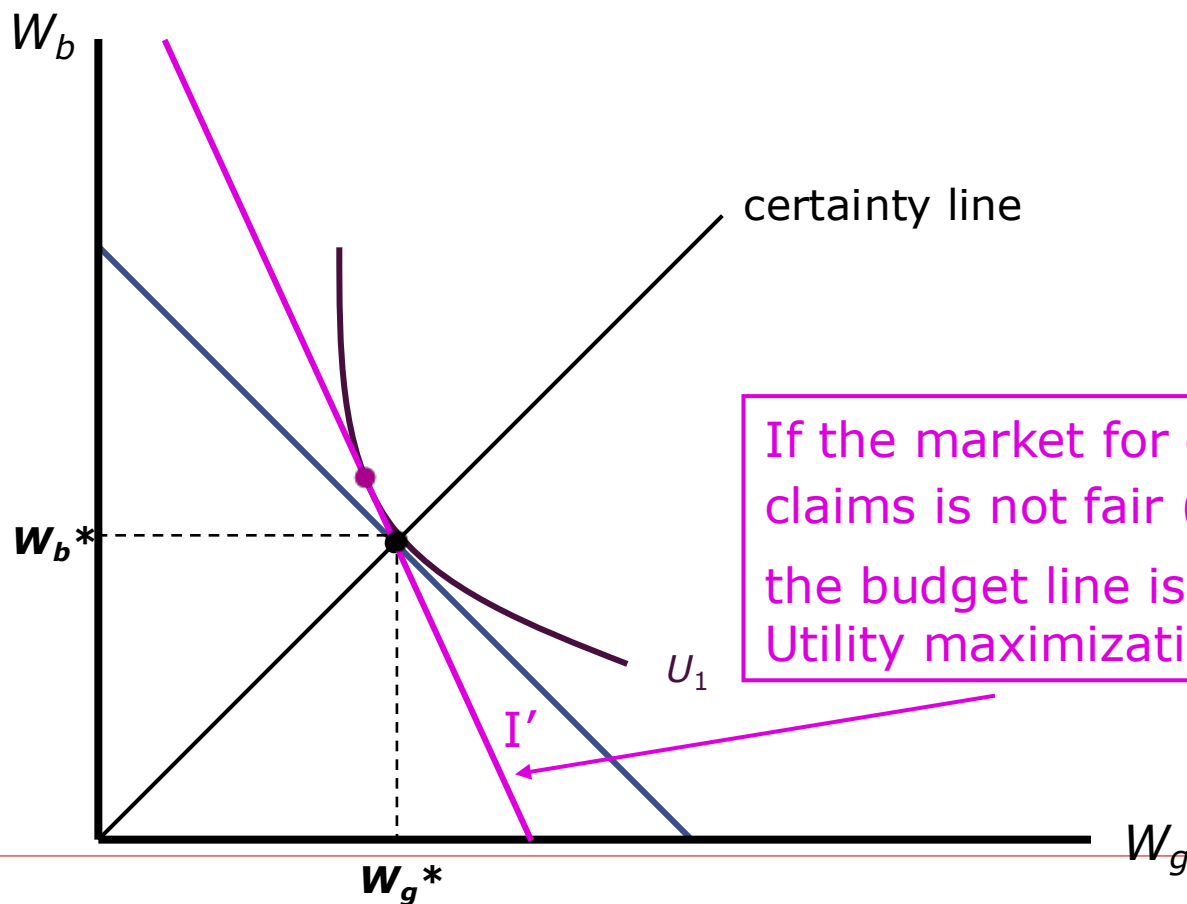
Risk Aversion

The individual maximizes utility on the certainty line where $W_g = W_b$

- W^* is independent of which state of the world occurs



Risk Aversion



If the market for contingent claims is not fair ($\frac{p_g}{p_b} > \frac{\pi}{1-\pi}$), the budget line is as I' , Utility maximization occur at $W_b > W_g$.

Risk Aversion and Risk Premiums

- Consider two people, each of whom starts with an initial wealth of W^*
- Each seeks to maximize an expected utility function of the form

$$V(W_g, W_b) = \pi \frac{W_g^R}{R} + (1 - \pi) \frac{W_b^R}{R}$$

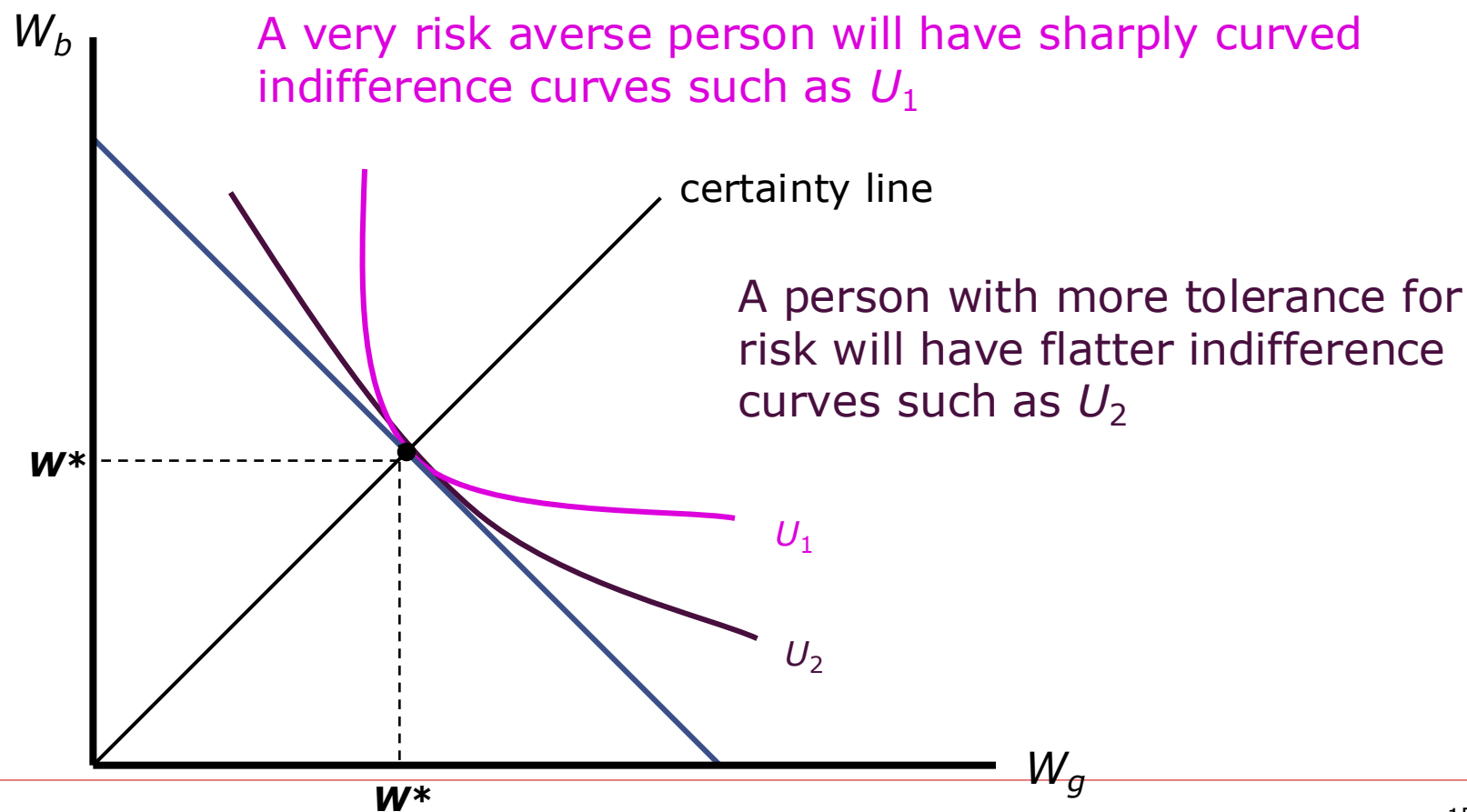
- This utility function exhibits constant relative risk aversion

Risk Aversion and Risk Premiums

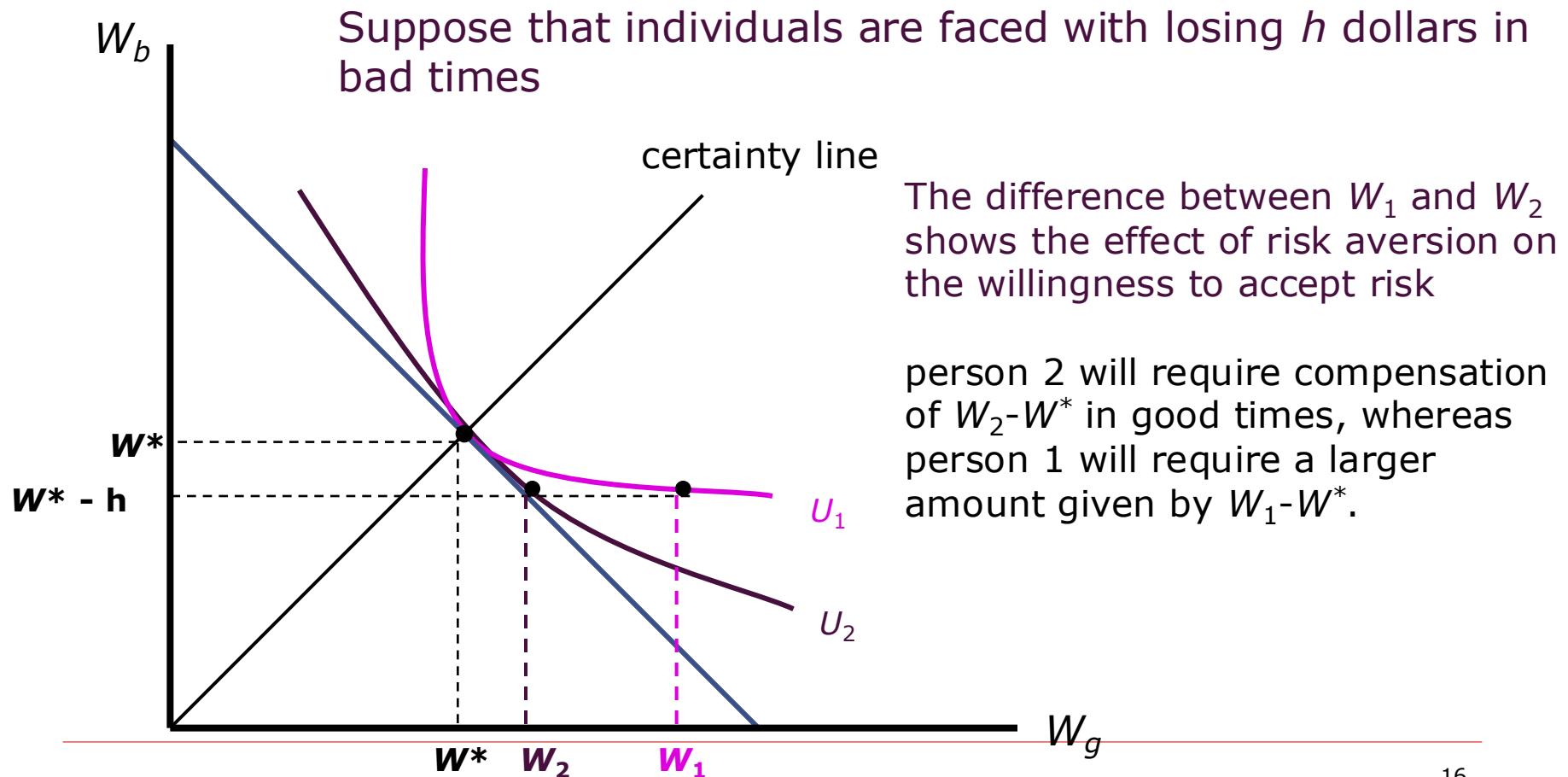
$$V(W_g, W_b) = \pi \frac{W_g^R}{R} + (1 - \pi) \frac{W_b^R}{R}$$

- The parameter R determines both the degree of risk aversion and the degree of curvature of indifference curves implied by the function
 - a very risk averse individual will have a large negative value for R

Risk Aversion and Risk Premiums



Risk Aversion and Risk Premiums



Insurance in the State-Preference Model

- Consider a person with wealth of \$100,000 who faces a 25% chance of losing his automobile worth \$20,000
 - wealth with no theft (W_g) = \$100,000 and probability of no theft = 0.75
 - wealth with a theft (W_b) = \$80,000 and probability of a theft = 0.25

Insurance in the State-Preference Model

- If we assume logarithmic utility, then

$$E(U) = 0.75U(W_g) + 0.25U(W_b)$$

$$E(U) = 0.75 \ln W_g + 0.25 \ln W_b$$

$$E(U) = 0.75 \ln (100,000) + 0.25 \ln (80,000)$$

$$E(U) = 11.45714$$

Insurance in the State-Preference Model

- The budget constraint is written in terms of the prices of the contingent commodities

$$p_g W_g^* + p_b W_b^* = p_g W_g + p_b W_b$$

- Assuming that these prices equal the probabilities of these two states

$$0.75(100,000) + 0.25(80,000) = 95,000$$

- The expected value of wealth = \$95,000

Insurance in the State-Preference Model

- The individual will move to the certainty line and receive an expected utility of

$$E(U) = \ln 95,000 = 11.46163$$

- to be able to do so, the individual must be able to transfer \$5,000 = -(95,000-100,000) in extra wealth in good times into \$15,000 = 95,000-80,000 of extra wealth in bad times
 - a fair insurance contract will allow this
 - the wealth changes promised by insurance (dW_b/dW_g)
= 15,000/-5,000 = -3

A Policy with a Deductible

- Suppose that the insurance policy costs \$4,900, but requires the person to incur the first \$1,000 of the loss

$$W_g = 100,000 - 4,900 = 95,100$$

$$W_b = 80,000 - 4,900 + 19,000 = 94,100$$

$$E(U) = 0.75 \ln 95,100 + 0.25 \ln 94,100$$

$$E(U) = 11.46004$$

- The policy still provides higher utility than doing nothing

The Demand for Insurance

- π = probability of accident,
 - W = income,
 - L = loss if the accident happened,
 - M = amount insured,
 - θ = premium percentage, then insurance premium = θM .
-
- There are two parties in the insurance market: insured (i.e., consumer) and insurer (i.e., insurance company)

The Demand for Insurance

□ Insured

- wealth at good times =
- wealth at bad times =
- The insured's objective is to choose M to maximize expected utility:
- To the insured, π, W, θ , and L are exogenous (i.e., given).
- To maximize utility, the first-order condition is

The Demand for Insurance

- Example: If $U(W) = \log W$, then the optimal M^* is given by

The Demand for Insurance

□ To the insurer,

■ revenue = θM ,

■ Cost = $\pi M + (1 - \pi)0 = \pi M$,

■ Hence, expected profit = $\theta M - \pi M$.

■ If insurance is actuarially fair (e.g., the insurance market is perfectly competitive), then the expected profit = 0, which implies $\theta = \pi$.

■ As a result, the premium percentage = probability of payout to the insured.

The Demand for Insurance

- Back to the insured,
- If the insurance is actuarially fair, then substituting $\theta = \pi$ into the first-order condition,
- One obtains

$$\begin{aligned}\pi(1 - \pi)U'(W - \pi M - L + M) - (1 - \pi)\pi U'(W - \pi M) &= 0 \\ U'(W - \pi M - L + M) &= U'(W - \pi M)\end{aligned}$$

- This equation means that the insured will achieve an optimum by choosing M to equalize the marginal utility of income across the two states (the accident state and the no-accident state).

The Demand for Insurance

- From mathematics, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$ if $f(x)$ is a monotonic function.
- Assuming $U'' < 0$, U' is thus monotonic, hence $W - \pi M - L + M = W - \pi M$, which implies that $M = L$. The insured is fully insured because $M = L$.

The Demand for Insurance

- From mathematics, $f(x_1) = f(x_2)$ implies that $x_1 = x_2$ if $f(x)$ is a monotonic function.
- Assuming $U'' < 0$, U' is thus monotonic, hence $W - \pi M - L + M = W - \pi M$, which implies that $M = L$. The insured is fully insured because $M = L$.
- As $\theta = \pi$ and $M = L$; the insured's utility = $\pi U(W - \theta M - L + M) + (1 - \pi)U(W - \theta M) = U(W - \pi L)$.
- Regardless of whether the accident occurs, the insured receives $U(W - \pi L)$ with certainty. Insurance essentially *removes* uncertainty!
- Example: If $U(W) = \log W$, and $\theta = \pi$, then $M^* = L$.

Methods for Reducing Uncertainty and Risk

- Four different methods that individuals can take to mitigate the problem of risk and uncertainty:
 - Insurance
 - Diversification
 - Flexibility
 - Information

Diversification

- Two firms, A and B.
 - Shares cost \$10.
 - With prob. $1/2$ A's profit is \$100 and B's profit is \$20.
 - With prob. $1/2$ A's profit is \$20 and B's profit is \$100.
 - You have \$100 to invest.
 - How?
-

Diversification

- ❑ (1) Buy only firm A's stock?
 - ❑ $\$100/10 = 10$ shares.
 - ❑ You earn \$1000 with prob. $1/2$ and \$200 with prob. $1/2$.
 - ❑ Expected earning: $\$500 + \$100 = \$600$
-

Diversification

- ❑ (2) Buy only firm B's stock?
 - ❑ $\$100/10 = 10$ shares.
 - ❑ You earn \$1000 with prob. $1/2$ and \$200 with prob. $1/2$.
 - ❑ Expected earning: $\$500 + \$100 = \$600$
-

Diversification

- ☐ (3) Buy 5 shares in each firm?
 - ☐ You earn \$600 for sure.
 - ☐ Diversification has maintained expected earning and lowered risk.
 - ☐ Typically, diversification lowers expected earnings in exchange for lowered risk.
-

Flexibility

□ Flexibility

- Allows the person to adjust the initial decision depending on how the future unfolds
- The more uncertain the future, the more valuable this flexibility
- Keeps the decision-maker from being tied to one course of action
 - And instead provides a number of options

Flexibility

- Financial option contract
 - Offers the right, but not the obligation, to buy or sell an asset
 - During some future period
 - At a certain price
- Real option
 - An option arising in a setting outside of financial markets

Flexibility

- All options share three fundamental attributes
 - Specify the underlying transaction
 - Specify a period over which the option may be exercised
 - Specifies a price

Flexibility

□ Model of real options

- Let x embody all the uncertainty in the economic environment
- The individual has some number, $i = 1, \dots, n$, of choices currently available
- $O_i(x)$ - payoffs provided by choice i
 - (x) allows each choice to provide a different pattern of returns depending on how the future turns out

Flexibility

□ Model of real options

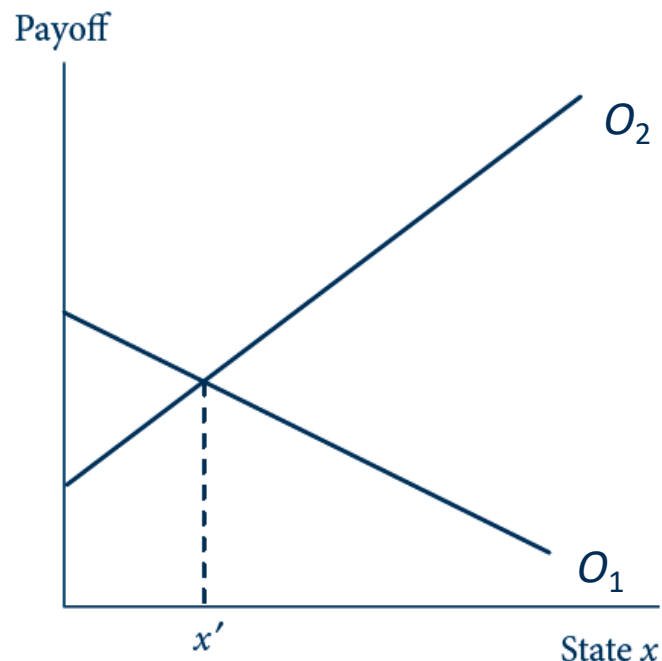
■ No flexibility

- Choose the single alternative that is best on average
- Expected utility from this choice: $\max\{E[U(O_1)], \dots, E[U(O_n)]\}$

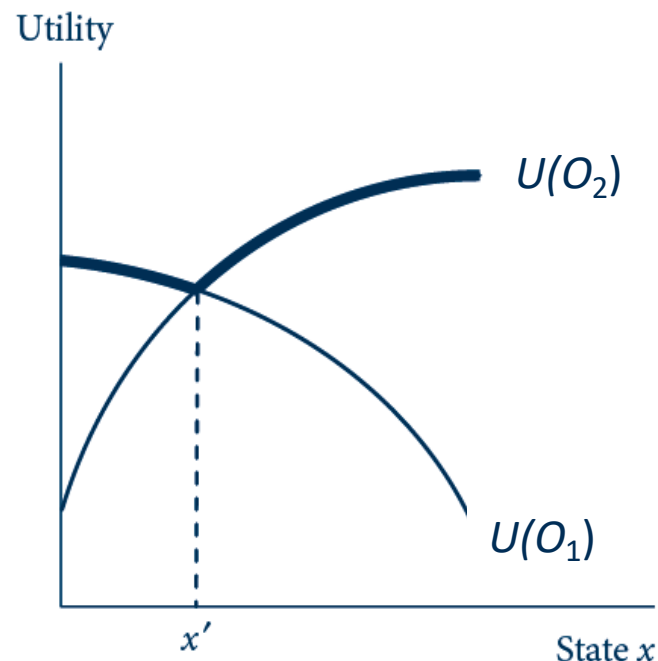
■ Flexibility

- Choose the best alternative
- Expected utility: $E\{\max[U(O_1), \dots, U(O_n)]\}$

The Nature of a Real Option



(a) Payoffs from alternatives



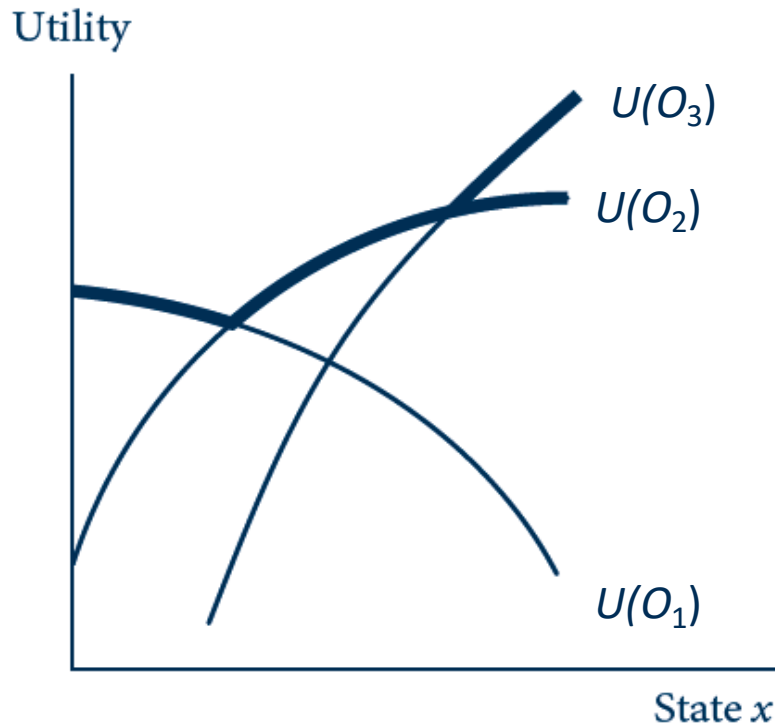
(b) Utilities from alternatives

- Panel (a) shows the payoffs and panel (b) shows the utilities provided by two alternatives across states of the world (x).
- If the decision has to be made upfront, the individual chooses the single curve having the highest expected utility.
- If the real option to make either decision can be preserved until later, the individual can obtain the expected utility of the upper envelope of the curves, shown in bold.

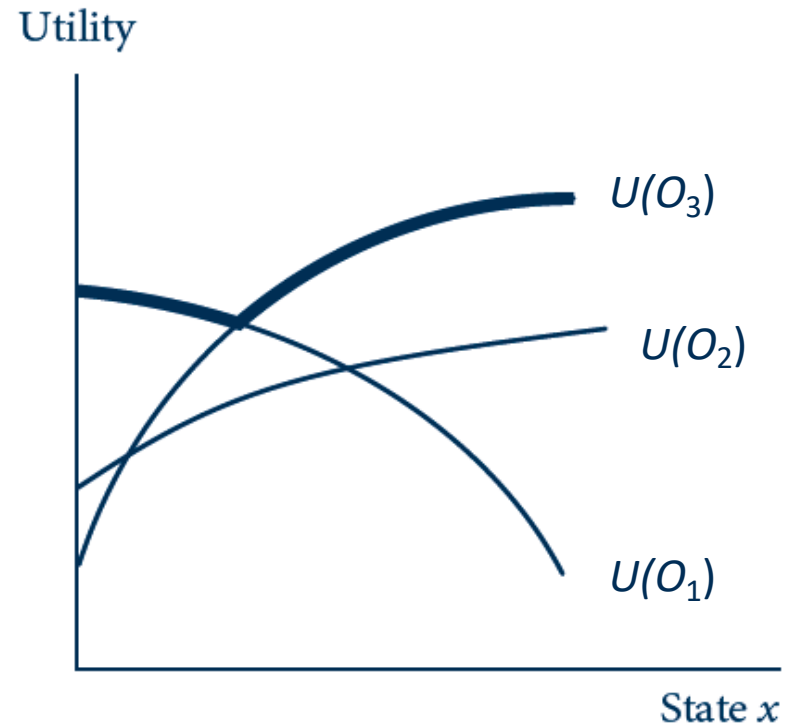
Flexibility

- More options are better (generally)
 - Options give the holder the right—but not the obligation—to choose them

More Options Cannot Make the Individual Decision-Maker Worse Off



(a) Additional valuable option



(b) Additional worthless option

- The addition of a third alternative to the two drawn in the previous figure is valuable in (a) because it shifts the upper envelope (shown in bold) of utilities up.
- The new alternative is worthless in (b) because it does not shift the upper envelope, but the individual is not worse off for having it.

Flexibility

□ Computing the value of a real option

- Let F be the fee that has to be paid

- For the ability to choose the best alternative *after* x has been realized instead of *before*

- The individual would be willing to pay the fee as long as:

$$E\{\max[U(O_1(x)-F), \dots, U(O_n(x)-F)]\} \geq \max\{E[U(O_1(x))], \dots, E[U(O_n(x))]\}$$

Practice example: Value of a Flexible-Fuel Car

□ $O_1(x) = 1 - x$

- The payoff from a fossil-fuel-only car

□ $O_2(x) = x$

- The payoff from a electric-only car

□ State of the world, x

- Reflects the relative importance of electricity compared with fossil fuels over the car's lifespan
- Random variable, uniformly distributed between 0 and 1

Value of a Flexible-Fuel Car

- Probability density function (PDF) is $f(x) = 1$
 - When the uniform random variable ranges between 0 and 1

- Suppose first that the car buyer is risk neutral
 - Utility level = payoff level
 - Forced to choose a electric vehicle
 - Expected utility:

$$E[O_2] = \int_0^1 O_2(x) f(x) dx = \frac{1}{2}$$

with $f(x) = 1$

Value of a Flexible-Fuel Car

□ Risk neutrality

- Flexible-fuel car is available
- Buyer: either $O_1(x)$ or $O_2(x)$, whichever is higher under the latter circumstances
- Expected utility:

$$\begin{aligned} E[\max(O_1, O_2)] &= \int_0^1 \max(1-x, x) f(x) dx = \\ &= \int_0^{1/2} (1-x) dx + \int_{1/2}^1 x dx = 3/4 \end{aligned}$$

- This individual is willing to pay 0.25 for the flexible-fuel car.

Value of a Flexible-Fuel Car

□ Risk aversion, $U(x) = \sqrt{x}$

■ Expected utility from an electric vehicle:

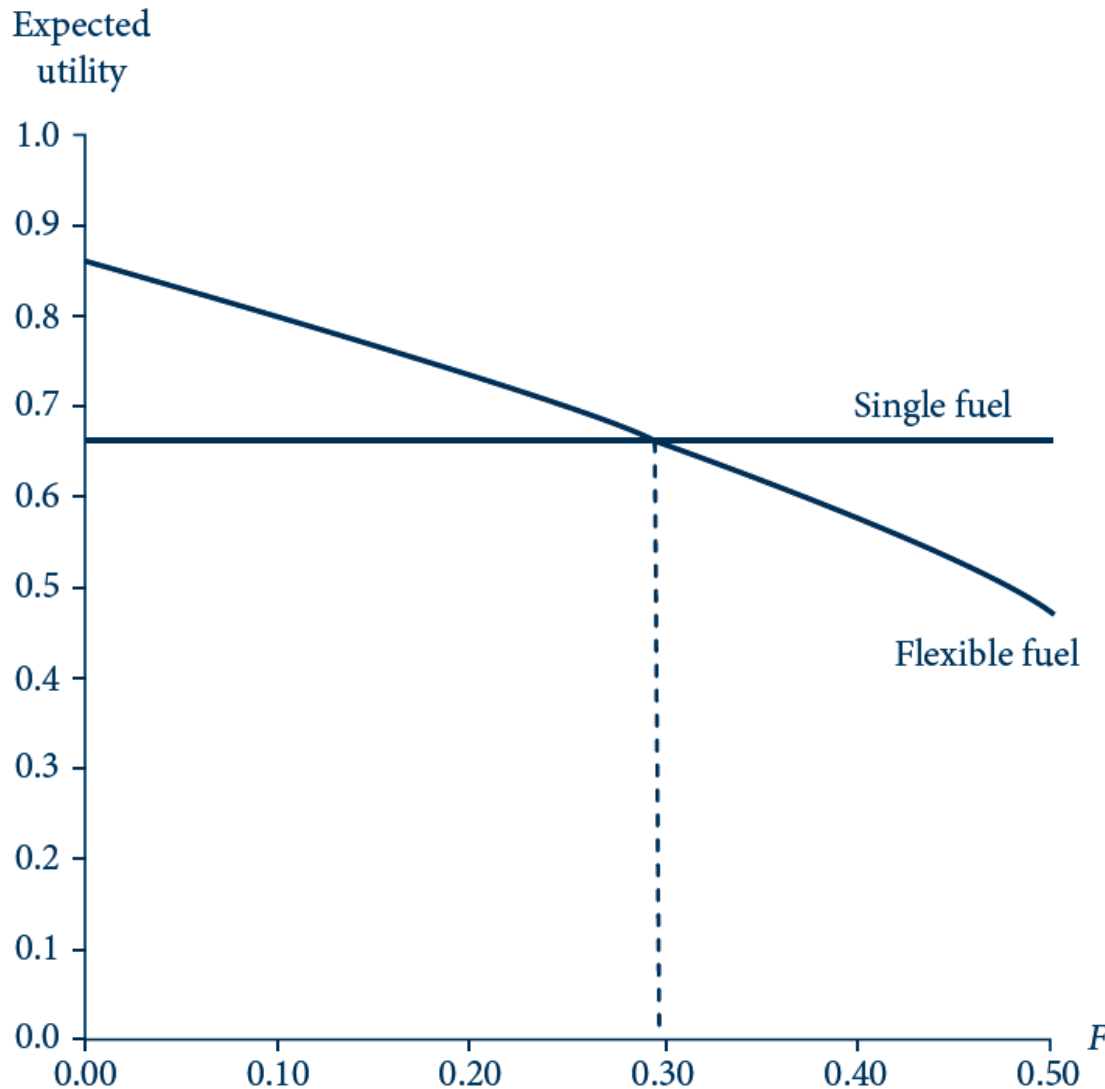
$$E[U(O_2)] = \int_0^1 \sqrt{O_2(x)} f(x) dx = \int_0^1 x^{1/2} dx = 2/3$$

■ Expected utility from a fossil-fuel car; $E[U(O_1)] = 2/3$

■ Expected utility from a flexible-fuel car that costs F more than a single-fuel car:

$$\begin{aligned} E\{\max[U(O_1(x) - F), U(O_2(x) - F)]\} &= \\ &= \int_0^1 \max(\sqrt{1-x-F}, \sqrt{x-F}) f(x) dx = \int_0^{1/2} \sqrt{1-x-F} dx + \int_{1/2}^1 \sqrt{x-F} dx \end{aligned}$$

Graphical Method for Computing the Premium for a Flexible-Fuel Car



To find the maximum premium F that the risk-averse buyer would be willing to pay for the flexible-fuel car, we plot the expected utility from a single-fuel car and from the flexible-fuel car, and see the value of F where the curves cross.

Value of a Flexible-Fuel Car

- we see that this value of F is slightly less than 0.3 (0.294 to be more precise).
- Therefore, the risk-averse buyer is willing to pay a premium of 0.294 for the flexible-fuel car, which is also the option value of this type of car.
- Scaling up by \$10,000 for more realistic monetary values, the price premium would be \$2,940.
- This is \$440 more than the risk-neutral buyer was willing to pay. Thus, the option value is greater in this case for the risk-averse buyer.